

27 Real-time clock (RTC)

27.1 Introduction

The RTC provides an automatic wake-up to manage all low-power modes.

The real-time clock (RTC) is an independent BCD timer/counter. The RTC provides a time-of-day clock/calendar with programmable alarm interrupts.

The RTC includes also a periodic programmable wake-up flag with interrupt capability.

Two 32-bit registers contain the seconds, minutes, hours (12- or 24-hour format), day (day of week), date (day of month), month, and year, expressed in binary coded decimal format (BCD). The sub-seconds value is also available in binary format.

Compensations for 28-, 29- (leap year), 30-, and 31-day months are performed automatically. Daylight saving time compensation can also be performed.

Additional 32-bit registers contain the programmable alarm subseconds, seconds, minutes, hours, day, and date.

A digital calibration feature is available to compensate for any deviation in crystal oscillator accuracy.

After Backup domain reset, all RTC registers are protected against possible parasitic write accesses.

As long as the supply voltage remains in the operating range, the RTC never stops, regardless of the device status (Run mode, low-power mode or under reset).

27.2 RTC main features

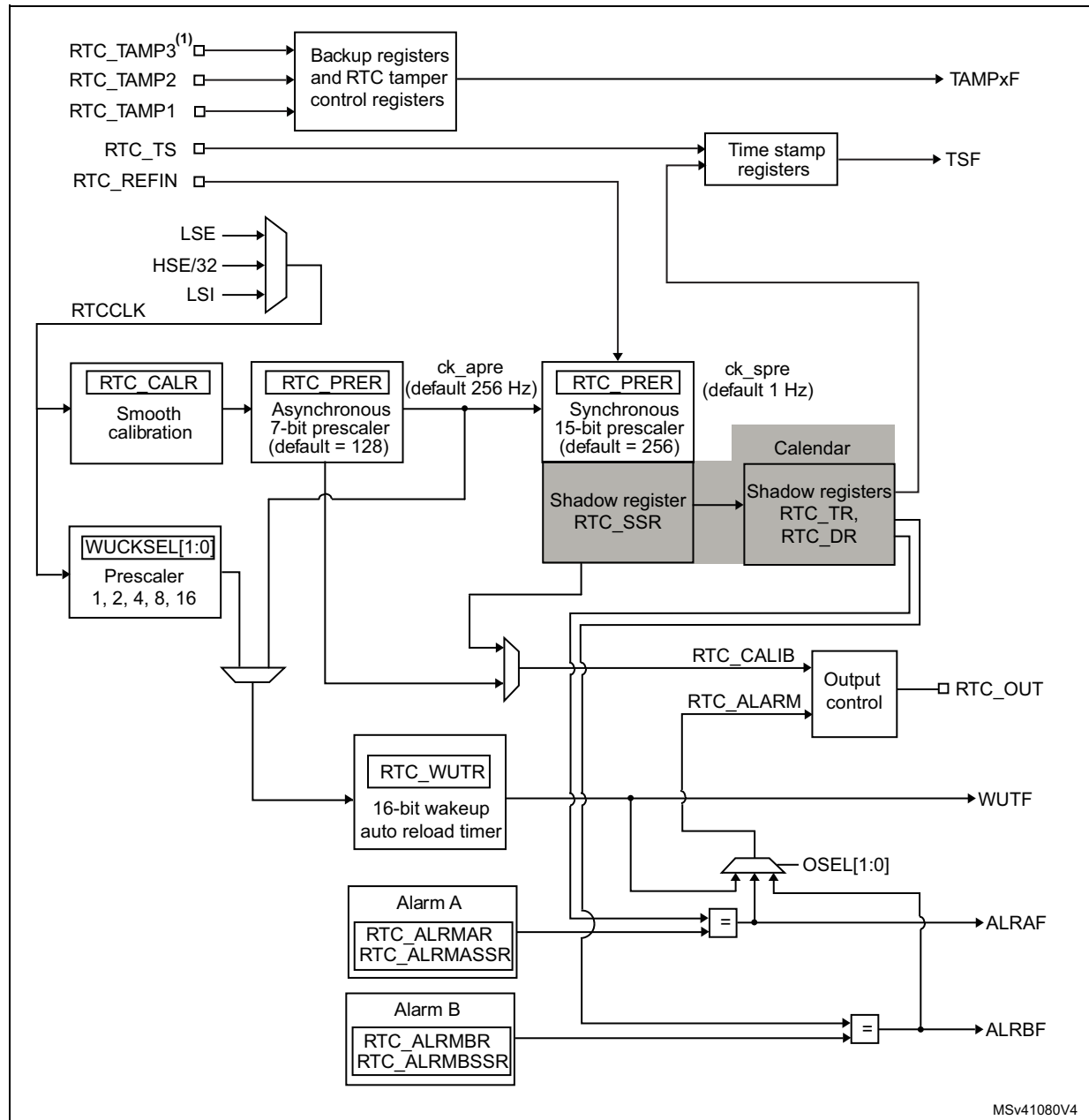
The RTC unit main features are the following (see [Figure 294: RTC block diagram](#)):

- Calendar with subseconds, seconds, minutes, hours (12 or 24 format), day (day of week), date (day of month), month, and year.
- Daylight saving compensation programmable by software.
- Programmable alarm with interrupt function. The alarm can be triggered by any combination of the calendar fields.
- Automatic wake-up unit generating a periodic flag that triggers an automatic wake-up interrupt.
- Reference clock detection: a more precise second source clock (50 or 60 Hz) can be used to enhance the calendar precision.
- Accurate synchronization with an external clock using the subsecond shift feature.
- Digital calibration circuit (periodic counter correction): 0.95 ppm accuracy, obtained in a calibration window of several seconds
- Time-stamp function for event saving
- Tamper detection event with configurable filter and internal pull-up
- Maskable interrupts/events:
 - Alarm A
 - Alarm B
 - Wake-up interrupt
 - Time-stamp
 - Tamper detection
- 16 backup registers.

27.3 RTC functional description

27.3.1 RTC block diagram

Figure 294. RTC block diagram



1. RTC_TAMP3 is not available on the STM32F303x6/8 and STM32F328 devices.

The RTC includes:

- Two alarms
- Three tamper events from I/Os
 - Tamper detection erases the backup registers.
- One timestamp event from I/O
- Tamper event detection can generate a timestamp event
- 16 x 32-bit backup registers on the STM32F303xB/C and 5 x 32 bit backup registers on the STM32F303x6/8
 - The backup registers (RTC_BKPxR) are implemented in the RTC domain that remains powered-on by VBAT when the VDD power is switched off.
- Output functions: RTC_OUT which selects one of the following two outputs:
 - RTC_CALIB: 512 Hz or 1Hz clock output (with an LSE frequency of 32.768 kHz). This output is enabled by setting the COE bit in the RTC_CR register.
 - RTC_ALARM: This output is enabled by configuring the OSEL[1:0] bits in the RTC_CR register which select the Alarm A, Alarm B or Wakeup outputs.
- Input functions:
 - RTC_TS: timestamp event
 - RTC_TAMP1: tamper1 event detection
 - RTC_TAMP2: tamper2 event detection
 - RTC_TAMP3: tamper3 event detection
 - RTC_REFIN: 50 or 60 Hz reference clock input

27.3.2 GPIOs controlled by the RTC

RTC_OUT, RTC_TS and RTC_TAMP1 are mapped on the same pin (PC13). PC13 pin configuration is controlled by the RTC, whatever the PC13 GPIO configuration, except for the RTC_ALARM output open-drain mode. In this particular case, the GPIO must be configured as input. The RTC functions mapped on PC13 are available in all low-power modes and in VBAT mode.

The selection of the RTC_ALARM output is performed through the RTC_TAFCR register as follows: the PC13VALUE bit is used to select whether the RTC_ALARM output is configured in push-pull or open drain mode.

When PC13 is not used as RTC function, it can be forced in output push-pull mode by setting the PC13MODE bit in the RTC_TAFCR. The output data value is then given by the PC13VALUE bit. In this case, PC13 output push-pull state and data are preserved in Standby mode.

The output mechanism follows the priority order shown in [Table 136](#).

When PC14 and PC15 are not used as LSE oscillator, they can be forced in output push-pull mode by setting the PC14MODE and PC15MODE bits in the RTC_TAFCR register respectively. The output data values are then given by PC14VALUE and PC15VALUE. In this case, the PC14 and PC15 output push-pull states and data values are preserved in Standby mode.

The output mechanism follows the priority order shown in [Table 137](#) and [Table 138](#).

Table 136. RTC pin PC13 configuration⁽¹⁾

Pin configuration and function	RTC_ALARM output enabled	RTC_CALIB output enabled	RTC_TAMP1 input enabled	RTC_TS input enabled	PC13MODE bit	PC13VALUE bit
RTC_ALARM output OD	1	Don't care	Don't care	Don't care	Don't care	0
RTC_ALARM output PP	1	Don't care	Don't care	Don't care	Don't care	1
RTC_CALIB output PP	0	1	Don't care	Don't care	Don't care	Don't care
RTC_TAMP1 input floating	0	0	1	0	Don't care	Don't care
RTC_TS and RTC_TAMP1 input floating	0	0	1	1	Don't care	Don't care
RTC_TS input floating	0	0	0	1	Don't care	Don't care
Output PP forced	0	0	0	0	1	PC13 output data value
Wake-up pin or Standard GPIO	0	0	0	0	0	Don't care

1. OD: open drain; PP: push-pull.

Table 137. LSE pin PC14 configuration⁽¹⁾

Pin configuration and function	LSEON bit in RCC_BDCR register	LSEBYP bit in RCC_BDCR register	PC14MODE bit	PC14VALUE bit
LSE oscillator	1	0	Don't care	Don't care
LSE bypass	1	1	Don't care	Don't care
Output PP forced	0	Don't care	1	PC14 output data value
Standard GPIO	0	Don't care	0	Don't care

1. OD: open drain; PP: push-pull.

Table 138. LSE pin PC15 configuration⁽¹⁾

Pin configuration and function	LSEON bit in RCC_BDCR register	LSEBYP bit in RCC_BDCR register	PC15MODE bit	PC15VALUE bit
LSE oscillator	1	0	Don't care	Don't care
Output PP forced	1	1	1	PC15 output data value
	0	Don't care		
Standard GPIO	0	Don't care	0	Don't care

1. OD: open drain; PP: push-pull.

27.3.3 Clock and prescalers

The RTC clock source (RTCCLK) is selected through the clock controller among the LSE clock, the LSI oscillator clock, and the HSE clock. For more information on the RTC clock source configuration, refer to [Section 9: Reset and clock control \(RCC\)](#).

A programmable prescaler stage generates a 1 Hz clock which is used to update the calendar. To minimize power consumption, the prescaler is split into 2 programmable prescalers (see [Figure 294: RTC block diagram](#)):

- A 7-bit asynchronous prescaler configured through the PREDIV_A bits of the RTC_PRER register.
- A 15-bit synchronous prescaler configured through the PREDIV_S bits of the RTC_PRER register.

Note: When both prescalers are used, it is recommended to configure the asynchronous prescaler to a high value to minimize consumption.

The asynchronous prescaler division factor is set to 128, and the synchronous division factor to 256, to obtain an internal clock frequency of 1 Hz (ck_spre) with an LSE frequency of 32.768 kHz.

The minimum division factor is 1 and the maximum division factor is 2^{22} .

This corresponds to a maximum input frequency of around 4 MHz.

f_{ck_apre} is given by the following formula:

$$f_{CK_APRE} = \frac{f_{RTCCLK}}{PREDIV_A + 1}$$

The ck_apre clock is used to clock the binary RTC_SSR subseconds downcounter. When it reaches 0, RTC_SSR is reloaded with the content of PREDIV_S.

f_{ck_spre} is given by the following formula:

$$f_{CK_SPRE} = \frac{f_{RTCCLK}}{(PREDIV_S + 1) \times (PREDIV_A + 1)}$$

The ck_spre clock can be used either to update the calendar or as timebase for the 16-bit wake-up auto-reload timer. To obtain short timeout periods, the 16-bit wake-up auto-reload timer can also run with the RTCCLK divided by the programmable 4-bit asynchronous prescaler (see [Section 27.3.6: Periodic auto-wake-up](#) for details).

27.3.4 Real-time clock and calendar

The RTC calendar time and date registers are accessed through shadow registers which are synchronized with PCLK (APB clock). They can also be accessed directly in order to avoid waiting for the synchronization duration.

- RTC_SSR for the subseconds
- RTC_TR for the time
- RTC_DR for the date

Every RTCCLK period, the current calendar value is copied into the shadow registers, and the RSF bit of RTC_ISR register is set (see [Section 27.6.4: RTC initialization and status](#)

[register \(RTC_ISR\)](#)). The copy is not performed in Stop and Standby mode. When exiting these modes, the shadow registers are updated after up to 1 RTCCLK period.

When the application reads the calendar registers, it accesses the content of the shadow registers. It is possible to make a direct access to the calendar registers by setting the BYPSHAD control bit in the RTC_CR register. By default, this bit is cleared, and the user accesses the shadow registers.

When reading the RTC_SSR, RTC_TR or RTC_DR registers in BYPSHAD=0 mode, the frequency of the APB clock (f_{APB}) must be at least 7 times the frequency of the RTC clock (f_{RTCCLK}).

The shadow registers are reset by system reset.

27.3.5 Programmable alarms

The RTC unit provides programmable alarm: Alarm A and Alarm B. The description below is given for Alarm A, but can be translated in the same way for Alarm B.

The programmable alarm function is enabled through the ALRAE bit in the RTC_CR register. The ALRAF is set to 1 if the calendar subseconds, seconds, minutes, hours, date or day match the values programmed in the alarm registers RTC_ALRMASR and RTC_ALRMAR. Each calendar field can be independently selected through the MSKx bits of the RTC_ALRMAR register, and through the MASKSSx bits of the RTC_ALRMASR register. The alarm interrupt is enabled through the ALRAIE bit in the RTC_CR register.

Caution: If the seconds field is selected (MSK1 bit reset in RTC_ALRMAR), the synchronous prescaler division factor set in the RTC_PRER register must be at least 3 to ensure correct behavior.

Alarm A and Alarm B (if enabled by bits OSEL[1:0] in RTC_CR register) can be routed to the RTC_ALARM output. RTC_ALARM output polarity can be configured through bit POL the RTC_CR register.

27.3.6 Periodic auto-wake-up

The periodic wake-up flag is generated by a 16-bit programmable auto-reload down-counter. The wake-up timer range can be extended to 17 bits.

The wake-up function is enabled through the WUTE bit in the RTC_CR register.

The wake-up timer clock input can be:

- RTC clock (RTCCLK) divided by 2, 4, 8, or 16.
When RTCCLK is LSE(32.768 kHz), this allows to configure the wake-up interrupt period from 122 μ s to 32 s, with a resolution down to 61 μ s.
- ck_spre (usually 1 Hz internal clock)
When ck_spre frequency is 1Hz, this allows to achieve a wake-up time from 1 s to around 36 hours with one-second resolution. This large programmable time range is divided in 2 parts:
 - from 1s to 18 hours when WUCKSEL [2:1] = 10
 - and from around 18h to 36h when WUCKSEL[2:1] = 11. In this last case 2¹⁶ is added to the 16-bit counter current value. When the initialization sequence is complete (see [Programming the wake-up timer on page 794](#)), the timer starts counting down. When the wake-up function is enabled, the down-counting remains active in low-power modes. In addition, when it reaches 0, the WUTF flag is set in

the RTC_ISR register, and the wake-up counter is automatically reloaded with its reload value (RTC_WUTR register value).

The WUTF flag must then be cleared by software.

When the periodic wake-up interrupt is enabled by setting the WUTIE bit in the RTC_CR register, it can exit the device from low-power modes.

The periodic wake-up flag can be routed to the RTC_ALARM output provided it has been enabled through bits OSEL[1:0] of RTC_CR register. RTC_ALARM output polarity can be configured through the POL bit in the RTC_CR register.

System reset, as well as low-power modes (Sleep, Stop and Standby) have no influence on the wake-up timer.

27.3.7 RTC initialization and configuration

RTC register access

The RTC registers are 32-bit registers. The APB interface introduces two wait states in RTC register accesses except on read accesses to calendar shadow registers when BYPSHAD = 0.

RTC register write protection

After system reset, the RTC registers are protected against parasitic write access by clearing the DBP bit in the PWR_CR register (refer to the power control section). DBP bit must be set in order to enable RTC registers write access.

After Backup domain reset, all the RTC registers are write-protected. Writing to the RTC registers is enabled by writing a key into the Write Protection register, RTC_WPR.

The following steps are required to unlock the write protection on all the RTC registers except for RTC_TAFCR, RTC_BKPxR and RTC_ISR[31:8].

1. Write '0xCA' into the RTC_WPR register.
2. Write '0x53' into the RTC_WPR register.

Writing a wrong key reactivates the write protection.

The protection mechanism is not affected by system reset.

Calendar initialization and configuration

To program the initial time and date calendar values, including the time format and the prescaler configuration, the following sequence is required:

1. Set INIT bit to 1 in the RTC_ISR register to enter initialization mode. In this mode, the calendar counter is stopped and its value can be updated.
2. Poll INITF bit of in the RTC_ISR register. The initialization phase mode is entered when INITF is set to 1. It takes around 2 RTCCLK clock cycles (due to clock synchronization).
3. To generate a 1 Hz clock for the calendar counter, program both the prescaler factors in RTC_PRER register.
4. Load the initial time and date values in the shadow registers (RTC_TR and RTC_DR), and configure the time format (12 or 24 hours) through the FMT bit in the RTC_CR register.
5. Exit the initialization mode by clearing the INIT bit. The actual calendar counter value is then automatically loaded and the counting restarts after 4 RTCCLK clock cycles.

When the initialization sequence is complete, the calendar starts counting.

Note: *After a system reset, the application can read the INITS flag in the RTC_ISR register to check if the calendar has been initialized or not. If this flag equals 0, the calendar has not been initialized since the year field is set at its Backup domain reset default value (0x00). To read the calendar after initialization, the software must first check that the RSF flag is set in the RTC_ISR register.*

Daylight saving time

The daylight saving time management is performed through bits SUB1H, ADD1H, and BKP of the RTC_CR register.

Using SUB1H or ADD1H, the software can subtract or add one hour to the calendar in one single operation without going through the initialization procedure.

In addition, the software can use the BKP bit to memorize this operation.

Programming the alarm

A similar procedure must be followed to program or update the programmable alarms. The procedure below is given for Alarm A but can be translated in the same way for Alarm B.

1. Clear ALRAE in RTC_CR to disable Alarm A.
2. Program the Alarm A registers (RTC_ALRMASR/RTC_ALRMAR).
3. Set ALRAE in the RTC_CR register to enable Alarm A again.

Note: *Each change of the RTC_CR register is taken into account after around 2 RTCCLK clock cycles due to clock synchronization.*

Programming the wake-up timer

The following sequence is required to configure or change the wake-up timer auto-reload value (WUT[15:0] in RTC_WUTR):

1. Clear WUTE in RTC_CR to disable the wake-up timer.
2. Poll WUTWF until it is set in RTC_ISR to make sure the access to wake-up auto-reload counter and to WUCKSEL[2:0] bits is allowed. It takes around 2 RTCCLK clock cycles (due to clock synchronization).
3. Program the wake-up auto-reload value WUT[15:0], and the wake-up clock selection (WUCKSEL[2:0] bits in RTC_CR). Set WUTE in RTC_CR to enable the timer again. The wake-up timer restarts down-counting. The WUTWF bit is cleared up to 2 RTCCLK clock cycles after WUTE is cleared, due to clock synchronization.

27.3.8 Reading the calendar

When BYPSHAD control bit is cleared in the RTC_CR register

To read the RTC calendar registers (RTC_SSR, RTC_TR and RTC_DR) properly, the APB1 clock frequency (f_{PCLK}) must be equal to or greater than seven times the RTC clock frequency (f_{RTCCLK}). This ensures a secure behavior of the synchronization mechanism.

If the APB1 clock frequency is less than seven times the RTC clock frequency, the software must read the calendar time and date registers twice. If the second read of the RTC_TR gives the same result as the first read, this ensures that the data is correct. Otherwise a third

read access must be done. In any case the APB1 clock frequency must never be lower than the RTC clock frequency.

The RSF bit is set in RTC_ISR register each time the calendar registers are copied into the RTC_SSR, RTC_TR and RTC_DR shadow registers. The copy is performed every RTCCLK cycle. To ensure consistency between the 3 values, reading either RTC_SSR or RTC_TR locks the values in the higher-order calendar shadow registers until RTC_DR is read. In case the software makes read accesses to the calendar in a time interval smaller than 1 RTCCLK period: RSF must be cleared by software after the first calendar read, and then the software must wait until RSF is set before reading again the RTC_SSR, RTC_TR and RTC_DR registers.

After waking up from low-power mode (Stop or Standby), RSF must be cleared by software. The software must then wait until it is set again before reading the RTC_SSR, RTC_TR and RTC_DR registers.

The RSF bit must be cleared after wake-up and not before entering low-power mode.

After a system reset, the software must wait until RSF is set before reading the RTC_SSR, RTC_TR and RTC_DR registers. Indeed, a system reset resets the shadow registers to their default values.

After an initialization (refer to [Calendar initialization and configuration on page 793](#)): the software must wait until RSF is set before reading the RTC_SSR, RTC_TR and RTC_DR registers.

After synchronization (refer to [Section 27.3.10: RTC synchronization](#)): the software must wait until RSF is set before reading the RTC_SSR, RTC_TR and RTC_DR registers.

When the BYPSHAD control bit is set in the RTC_CR register (bypass shadow registers)

Reading the calendar registers gives the values from the calendar counters directly, thus eliminating the need to wait for the RSF bit to be set. This is especially useful after exiting from low-power modes (STOP or Standby), since the shadow registers are not updated during these modes.

When the BYPSHAD bit is set to 1, the results of the different registers might not be coherent with each other if an RTCCLK edge occurs between two read accesses to the registers. Additionally, the value of one of the registers may be incorrect if an RTCCLK edge occurs during the read operation. The software must read all the registers twice, and then compare the results to confirm that the data is coherent and correct. Alternatively, the software can just compare the two results of the least-significant calendar register.

Note: While BYPSHAD=1, instructions which read the calendar registers require one extra APB cycle to complete.

27.3.9 Resetting the RTC

The calendar shadow registers (RTC_SSR, RTC_TR and RTC_DR) and some bits of the RTC status register (RTC_ISR) are reset to their default values by all available system reset sources.

On the contrary, the following registers are reset to their default values by a Backup domain reset and are not affected by a system reset: the RTC current calendar registers, the RTC control register (RTC_CR), the prescaler register (RTC_PRER), the RTC calibration register (RTC_CALR), the RTC shift register (RTC_SHIFTR), the RTC timestamp registers

(RTC_TSSSR, RTC_TSTR and RTC_TSDR), the RTC tamper and alternate function configuration register (RTC_TAFCR), the RTC backup registers (RTC_BKPxR), the wake-up timer register (RTC_WUTR), the Alarm A and Alarm B registers (RTC_ALRMASSR/RTC_ALRMAR and RTC_ALRMBSSR/RTC_ALRMBR).

In addition, when it is clocked by the LSE, the RTC keeps on running under system reset if the reset source is different from the Backup domain reset one (refer to the RTC clock section of the Reset and clock controller for details on the list of RTC clock sources not affected by system reset). When a Backup domain reset occurs, the RTC is stopped and all the RTC registers are set to their reset values.

27.3.10 RTC synchronization

The RTC can be synchronized to a remote clock with a high degree of precision. After reading the sub-second field (RTC_SSR or RTC_TSSSR), a calculation can be made of the precise offset between the times being maintained by the remote clock and the RTC. The RTC can then be adjusted to eliminate this offset by “shifting” its clock by a fraction of a second using RTC_SHIFTR.

RTC_SSR contains the value of the synchronous prescaler counter. This allows one to calculate the exact time being maintained by the RTC down to a resolution of $1 / (\text{PREDIV_S} + 1)$ seconds. As a consequence, the resolution can be improved by increasing the synchronous prescaler value (PREDIV_S[14:0]). The maximum resolution allowed (30.52 μ s with a 32768 Hz clock) is obtained with PREDIV_S set to 0x7FFF.

However, increasing PREDIV_S means that PREDIV_A must be decreased in order to maintain the synchronous prescaler output at 1 Hz. In this way, the frequency of the asynchronous prescaler output increases, which may increase the RTC dynamic consumption.

The RTC can be finely adjusted using the RTC shift control register (RTC_SHIFTR). Writing to RTC_SHIFTR can shift (either delay or advance) the clock by up to a second with a resolution of $1 / (\text{PREDIV_S} + 1)$ seconds. The shift operation consists of adding the SUBFS[14:0] value to the synchronous prescaler counter SS[15:0]: this will delay the clock. If at the same time the ADD1S bit is set, this results in adding one second and at the same time subtracting a fraction of second, so this will advance the clock.

Caution: Before initiating a shift operation, the user must check that SS[15] = 0 in order to ensure that no overflow will occur.

As soon as a shift operation is initiated by a write to the RTC_SHIFTR register, the SHPF flag is set by hardware to indicate that a shift operation is pending. This bit is cleared by hardware as soon as the shift operation has completed.

Caution: This synchronization feature is not compatible with the reference clock detection feature: firmware must not write to RTC_SHIFTR when REFCKON=1.

27.3.11 RTC reference clock detection

The update of the RTC calendar can be synchronized to a reference clock, RTC_REFIN, which is usually the mains frequency (50 or 60 Hz). The precision of the RTC_REFIN reference clock should be higher than the 32.768 kHz LSE clock. When the RTC_REFIN detection is enabled (REFCKON bit of RTC_CR set to 1), the calendar is still clocked by the LSE, and RTC_REFIN is used to compensate for the imprecision of the calendar update frequency (1 Hz).

Each 1 Hz clock edge is compared to the nearest RTC_REFIN clock edge (if one is found within a given time window). In most cases, the two clock edges are properly aligned. When the 1 Hz clock becomes misaligned due to the imprecision of the LSE clock, the RTC shifts the 1 Hz clock a bit so that future 1 Hz clock edges are aligned. Thanks to this mechanism, the calendar becomes as precise as the reference clock.

The RTC detects if the reference clock source is present by using the 256 Hz clock (ck_apre) generated from the 32.768 kHz quartz. The detection is performed during a time window around each of the calendar updates (every 1 s). The window equals 7 ck_apre periods when detecting the first reference clock edge. A smaller window of 3 ck_apre periods is used for subsequent calendar updates.

Each time the reference clock is detected in the window, the synchronous prescaler which outputs the ck_spre clock is forced to reload. This has no effect when the reference clock and the 1 Hz clock are aligned because the prescaler is being reloaded at the same moment. When the clocks are not aligned, the reload shifts future 1 Hz clock edges a little for them to be aligned with the reference clock.

If the reference clock halts (no reference clock edge occurred during the 3 ck_apre window), the calendar is updated continuously based solely on the LSE clock. The RTC then waits for the reference clock using a large 7 ck_apre period detection window centered on the ck_spre edge.

When the RTC_REFIN detection is enabled, PREDIV_A and PREDIV_S must be set to their default values:

- PREDIV_A = 0x007F
- PREDIV_S = 0x00FF

Note: RTC_REFIN clock detection is not available in Standby mode.

27.3.12 RTC smooth digital calibration

The RTC frequency can be digitally calibrated with a resolution of about 0.954 ppm with a range from -487.1 ppm to +488.5 ppm. The correction of the frequency is performed using series of small adjustments (adding and/or subtracting individual RTCCLK pulses). These adjustments are fairly well distributed so that the RTC is well calibrated even when observed over short durations of time.

The smooth digital calibration is performed during a cycle of about 2^{20} RTCCLK pulses, or 32 seconds when the input frequency is 32768 Hz. This cycle is maintained by a 20-bit counter, cal_cnt[19:0], clocked by RTCCLK.

The smooth calibration register (RTC_CALR) specifies the number of RTCCLK clock cycles to be masked during the 32-second cycle:

- Setting the bit CALM[0] to 1 causes exactly one pulse to be masked during the 32-second cycle.
- Setting CALM[1] to 1 causes two additional cycles to be masked
- Setting CALM[2] to 1 causes four additional cycles to be masked
- and so on up to CALM[8] set to 1 which causes 256 clocks to be masked.

Note: CALM[8:0] (RTC_CALR) specifies the number of RTCCLK pulses to be masked during the 32-second cycle. Setting the bit CALM[0] to '1' causes exactly one pulse to be masked during the 32-second cycle at the moment when cal_cnt[19:0] is 0x80000; CALM[1]=1 causes two other cycles to be masked (when cal_cnt is 0x40000 and 0xC0000); CALM[2]=1

causes four other cycles to be masked (cal_cnt = 0x20000/0x60000/0xA0000/ 0xE0000); and so on up to CALM[8]=1 which causes 256 clocks to be masked (cal_cnt = 0xFF800).

While CALM allows the RTC frequency to be reduced by up to 487.1 ppm with fine resolution, the bit CALP can be used to increase the frequency by 488.5 ppm. Setting CALP to '1' effectively inserts an extra RTCCLK pulse every 2^{11} RTCCLK cycles, which means that 512 clocks are added during every 32-second cycle.

Using CALM together with CALP, an offset ranging from -511 to +512 RTCCLK cycles can be added during the 32-second cycle, which translates to a calibration range of -487.1 ppm to +488.5 ppm with a resolution of about 0.954 ppm.

The formula to calculate the effective calibrated frequency (F_{CAL}) given the input frequency (F_{RTCCLK}) is as follows:

$$F_{CAL} = F_{RTCCLK} \times [1 + (CALP \times 512 - CALM) / (2^{20} + CALM - CALP \times 512)]$$

Calibration when PREDIV_A < 3

The CALP bit can not be set to 1 when the asynchronous prescaler value (PREDIV_A bits in RTC_PRER register) is less than 3. If CALP was already set to 1 and PREDIV_A bits are set to a value less than 3, CALP is ignored and the calibration operates as if CALP was equal to 0.

To perform a calibration with PREDIV_A less than 3, the synchronous prescaler value (PREDIV_S) should be reduced so that each second is accelerated by 8 RTCCLK clock cycles, which is equivalent to adding 256 clock cycles every 32 seconds. As a result, between 255 and 256 clock pulses (corresponding to a calibration range from 243.3 to 244.1 ppm) can effectively be added during each 32-second cycle using only the CALM bits.

With a nominal RTCCLK frequency of 32768 Hz, when PREDIV_A equals 1 (division factor of 2), PREDIV_S should be set to 16379 rather than 16383 (4 less). The only other interesting case is when PREDIV_A equals 0, PREDIV_S should be set to 32759 rather than 32767 (8 less).

If PREDIV_S is reduced in this way, the formula given the effective frequency of the calibrated input clock is as follows:

$$F_{CAL} = F_{RTCCLK} \times [1 + (256 - CALM) / (2^{20} + CALM - 256)]$$

In this case, CALM[7:0] equals 0x100 (the midpoint of the CALM range) is the correct setting if RTCCLK is exactly 32768.00 Hz.

Verifying the RTC calibration

RTC precision is ensured by measuring the precise frequency of RTCCLK and calculating the correct CALM value and CALP values. An optional 1 Hz output is provided to allow applications to measure and verify the RTC precision.

Measuring the precise frequency of the RTC over a limited interval can result in a measurement error of up to 2 RTCCLK clock cycles over the measurement period, depending on how the digital calibration cycle is aligned with the measurement period.

However, this measurement error can be eliminated if the measurement period is the same length as the calibration cycle period. In this case, the only error observed is the error due to the resolution of the digital calibration.

- By default, the calibration cycle period is 32 seconds.

Using this mode and measuring the accuracy of the 1 Hz output over exactly 32 seconds guarantees that the measure is within 0.477 ppm (0.5 RTCCLK cycles over 32 seconds, due to the limitation of the calibration resolution).

- CALW16 bit of the RTC_CALR register can be set to 1 to force a 16- second calibration cycle period.

In this case, the RTC precision can be measured during 16 seconds with a maximum error of 0.954 ppm (0.5 RTCCLK cycles over 16 seconds). However, since the calibration resolution is reduced, the long term RTC precision is also reduced to 0.954 ppm: CALM[0] bit is stuck at 0 when CALW16 is set to 1.

- CALW8 bit of the RTC_CALR register can be set to 1 to force a 8- second calibration cycle period.

In this case, the RTC precision can be measured during 8 seconds with a maximum error of 1.907 ppm (0.5 RTCCLK cycles over 8s). The long term RTC precision is also reduced to 1.907 ppm: CALM[1:0] bits are stuck at 00 when CALW8 is set to 1.

Re-calibration on-the-fly

The calibration register (RTC_CALR) can be updated on-the-fly while RTC_ISR/INITF=0, by using the follow process:

1. Poll the RTC_ISR/RECALPF (re-calibration pending flag).
2. If it is set to 0, write a new value to RTC_CALR, if necessary. RECALPF is then automatically set to 1
3. Within three ck_apre cycles after the write operation to RTC_CALR, the new calibration settings take effect.

27.3.13 Time-stamp function

Time-stamp is enabled by setting the TSE bit of RTC_CR register to 1.

The calendar is saved in the time-stamp registers (RTC_TSSSR, RTC_TSTR, RTC_TSDR) when a time-stamp event is detected on the RTC_TS pin.

When a time-stamp event occurs, the time-stamp flag bit (TSF) in RTC_ISR register is set.

By setting the TSIE bit in the RTC_CR register, an interrupt is generated when a time-stamp event occurs.

If a new time-stamp event is detected while the time-stamp flag (TSF) is already set, the time-stamp overflow flag (TSOVF) flag is set and the time-stamp registers (RTC_TSTR and RTC_TSDR) maintain the results of the previous event.

Note: *TSF is set 2 ck_apre cycles after the time-stamp event occurs due to synchronization process.*

There is no delay in the setting of TSOVF. This means that if two time-stamp events are close together, TSOVF can be seen as '1' while TSF is still '0'. As a consequence, it is recommended to poll TSOVF only after TSF has been set.

Caution: If a time-stamp event occurs immediately after the TSF bit is supposed to be cleared, then both TSF and TSOVF bits are set. To avoid masking a time-stamp event occurring at the same moment, the application must not write '0' into TSF bit unless it has already read it to '1'.

Optionally, a tamper event can cause a time-stamp to be recorded. See the description of the TAMPTS control bit in [Section 27.6.16: RTC tamper and alternate function configuration register \(RTC_TAFCR\)](#).

27.3.14 Tamper detection

The RTC_TAMPx input events can be configured either for edge detection, or for level detection with filtering.

The tamper detection can be configured for the following purposes:

- erase the RTC backup registers
- generate an interrupt, capable to wake-up from Stop and Standby modes

RTC backup registers

The backup registers (RTC_BKPxR) are not reset by system reset or when the device wakes up from Standby mode.

The backup registers are reset when a tamper detection event occurs (see [Section 27.6.19: RTC backup registers \(RTC_BKPxR\)](#) and [Tamper detection initialization on page 800](#)).

Tamper detection initialization

Each input can be enabled by setting the corresponding TAMPxE bits to 1 in the RTC_TAFCR register.

Each RTC_TAMPx tamper detection input is associated with a flag TAMPxF in the RTC_ISR register.

The TAMPxF flag is asserted after the tamper event on the pin, with the latency provided below:

- 3 ck_apre cycles when TAMPFLT differs from 0x0 (Level detection with filtering)
- 3 ck_apre cycles when TAMPTS=1 (Timestamp on tamper event)
- No latency when TAMPFLT=0x0 (Edge detection) and TAMPTS=0

A new tamper occurring on the same pin during this period and as long as TAMPxF is set cannot be detected.

By setting the TAMPIE bit in the RTC_TAFCR register, an interrupt is generated when a tamper detection event occurs. .

Timestamp on tamper event

With TAMPTS set to '1', any tamper event causes a timestamp to occur. In this case, either the TSF bit or the TSOVF bit are set in RTC_ISR, in the same manner as if a normal timestamp event occurs. The affected tamper flag register TAMPxF is set at the same time that TSF or TSOVF is set.

Edge detection on tamper inputs

If the TAMPFLT bits are "00", the RTC_TAMPx pins generate tamper detection events when either a rising edge or a falling edge is observed depending on the corresponding TAMPxTRG bit. The internal pull-up resistors on the RTC_TAMPx inputs are deactivated when edge detection is selected.

- Caution:** To avoid losing tamper detection events, the signal used for edge detection is logically ANDed with the corresponding TAMPxE bit in order to detect a tamper detection event in case it occurs before the RTC_TAMPx pin is enabled.
- When TAMPxTRG = 0: if the RTC_TAMPx is already high before tamper detection is enabled (TAMPxE bit set to 1), a tamper event is detected as soon as the RTC_TAMPx input is enabled, even if there was no rising edge on the RTC_TAMPx input after TAMPxE was set.
 - When TAMPxTRG = 1: if the RTC_TAMPx is already low before tamper detection is enabled, a tamper event is detected as soon as the RTC_TAMPx input is enabled (even if there was no falling edge on the RTC_TAMPx input after TAMPxE was set).

After a tamper event has been detected and cleared, the RTC_TAMPx should be disabled and then re-enabled (TAMPxE set to 1) before re-programming the backup registers (RTC_BKPxR). This prevents the application from writing to the backup registers while the RTC_TAMPx input value still indicates a tamper detection. This is equivalent to a level detection on the RTC_TAMPx input.

Note: *Tamper detection is still active when V_{DD} power is switched off. To avoid unwanted resetting of the backup registers, the pin to which the RTC_TAMPx is mapped should be externally tied to the correct level.*

Level detection with filtering on RTC_TAMPx inputs

Level detection with filtering is performed by setting TAMPFLT to a non-zero value. A tamper detection event is generated when either 2, 4, or 8 (depending on TAMPFLT) consecutive samples are observed at the level designated by the TAMPxTRG bits.

The RTC_TAMPx inputs are precharged through the I/O internal pull-up resistance before its state is sampled, unless disabled by setting TAMPPUDIS to 1. The duration of the precharge is determined by the TAMPPRCH bits, allowing for larger capacitances on the RTC_TAMPx inputs.

The trade-off between tamper detection latency and power consumption through the pull-up can be optimized by using TAMPFREQ to determine the frequency of the sampling for level detection.

Note: *Refer to the datasheets for the electrical characteristics of the pull-up resistors.*

27.3.15 Calibration clock output

When the COE bit is set to 1 in the RTC_CR register, a reference clock is provided on the RTC_CALIB device output.

If the COSEL bit in the RTC_CR register is reset and PREDIV_A = 0x7F, the RTC_CALIB frequency is $f_{\text{RTCCLK}}/64$. This corresponds to a calibration output at 512 Hz for an RTCCLK frequency at 32.768 kHz. The RTC_CALIB duty cycle is irregular: there is a light jitter on falling edges. It is therefore recommended to use rising edges.

When COSEL is set and "PREDIV_S+1" is a non-zero multiple of 256 (i.e: PREDIV_S[7:0] = 0xFF), the RTC_CALIB frequency is $f_{\text{RTCCLK}}/(256 * (\text{PREDIV_A}+1))$. This corresponds to a calibration output at 1 Hz for prescaler default values (PREDIV_A = 0x7F, PREDIV_S = 0xFF), with an RTCCLK frequency at 32.768 kHz. The 1 Hz output is affected when a shift operation is on going and may toggle during the shift operation (SHPF=1).

Note: When COSEL bit is cleared, the RTC_CALIB output is the output of the 6th stage of the asynchronous prescaler.

When COSEL bit is set, the RTC_CALIB output is the output of the 8th stage of the synchronous prescaler.

27.3.16 Alarm output

The OSEL[1:0] control bits in the RTC_CR register are used to activate the alarm output RTC_ALARM, and to select the function which is output. These functions reflect the contents of the corresponding flags in the RTC_ISR register.

The polarity of the output is determined by the POL control bit in RTC_CR so that the opposite of the selected flag bit is output when POL is set to 1.

Alarm output

The RTC_ALARM pin can be configured in output open drain or output push-pull using RTC_TAFCR register.

Note: Once the RTC_ALARM output is enabled, it has priority over RTC_CALIB (COE bit is don't care and must be kept cleared).

27.4 RTC low-power modes

Table 139. Effect of low-power modes on RTC

Mode	Description
Sleep	No effect RTC interrupts cause the device to exit the Sleep mode.
Stop	The RTC remains active when the RTC clock source is LSE or LSI. RTC alarm, RTC tamper event, RTC timestamp event, and RTC wake-up cause the device to exit the Stop mode.
Standby	The RTC remains active when the RTC clock source is LSE or LSI. RTC alarm, RTC tamper event, RTC timestamp event, and RTC wake-up cause the device to exit the Standby mode.

27.5 RTC interrupts

All RTC interrupts are connected to the EXTI controller. Refer to [Section 14.3: EXTI registers](#).

To enable the RTC Alarm interrupt, the following sequence is required:

1. Configure and enable the EXTI line corresponding to the RTC Alarm event in interrupt mode and select the rising edge sensitivity.
2. Configure and enable the RTC_ALARM IRQ channel in the NVIC.
3. Configure the RTC to generate RTC alarms.

To enable the RTC Tamper interrupt, the following sequence is required:

1. Configure and enable the EXTI line corresponding to the RTC Tamper event in interrupt mode and select the rising edge sensitivity.
2. Configure and Enable the RTC_TAMP_STAMP IRQ channel in the NVIC.
3. Configure the RTC to detect the RTC tamper event.

To enable the RTC TimeStamp interrupt, the following sequence is required:

1. Configure and enable the EXTI line corresponding to the RTC TimeStamp event in interrupt mode and select the rising edge sensitivity.
2. Configure and Enable the RTC_TAMP_STAMP IRQ channel in the NVIC.
3. Configure the RTC to detect the RTC time-stamp event.

To enable the Wakeup timer interrupt, the following sequence is required:

1. Configure and enable the EXTI line corresponding to the Wakeup timer even in interrupt mode and select the rising edge sensitivity.
2. Configure and Enable the RTC_WKUP IRQ channel in the NVIC.
3. Configure the RTC to detect the RTC Wakeup timer event.

Table 140. Interrupt control bits

Interrupt event	Event flag	Enable control bit	Exit from Sleep mode	Exit from Stop mode	Exit from Standby mode
Alarm A	ALRAF	ALRAIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾
Alarm B	ALRBF	ALRBIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾
RTC_TS input (timestamp)	TSF	TSIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾
RTC_TAMP1 input detection	TAMP1F	TAMPIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾
RTC_TAMP2 input detection	TAMP2F	TAMPIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾
RTC_TAMP3 input detection	TAMP3F	TAMPIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾
Wakeup timer interrupt	WUTF	WUTIE	Yes	Yes ⁽¹⁾	Yes ⁽¹⁾

1. Wakeup from STOP and Standby modes is possible only when the RTC clock source is LSE or LSI.

27.6 RTC registers

Refer to [Section 2.2 on page 47](#) of the reference manual for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by words (32-bit).

27.6.1 RTC time register (RTC_TR)

The RTC_TR is the calendar time shadow register. This register must be written in initialization mode only. Refer to [Calendar initialization and configuration on page 793](#) and [Reading the calendar on page 794](#).

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x00

Backup domain reset value: 0x0000 0000

System reset: 0x0000 0000 when BYPSHAD = 0. Not affected when BYPSHAD = 1.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]			
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]			
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bit 22 **PM**: AM/PM notation

0: AM or 24-hour format

1: PM

Bits 21:20 **HT[1:0]**: Hour tens in BCD format

Bits 19:16 **HU[3:0]**: Hour units in BCD format

Bit 15 Reserved, must be kept at reset value.

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format

Bits 11:8 **MNU[3:0]**: Minute units in BCD format

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **ST[2:0]**: Second tens in BCD format

Bits 3:0 **SU[3:0]**: Second units in BCD format

27.6.2 RTC date register (RTC_DR)

The RTC_DR is the calendar date shadow register. This register must be written in initialization mode only. Refer to [Calendar initialization and configuration on page 793](#) and [Reading the calendar on page 794](#).

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x04

Backup domain reset value: 0x0000 2101

System reset: 0x0000 2101 when BYPSHAD = 0. Not affected when BYPSHAD = 1.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	YT[3:0]				YU[3:0]			
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDU[2:0]			MT	MU[3:0]				Res.	Res.	DT[1:0]		DU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:20 **YT[3:0]**: Year tens in BCD format

Bits 19:16 **YU[3:0]**: Year units in BCD format

Bits 15:13 **WDU[2:0]**: Week day units

000: forbidden

001: Monday

...

111: Sunday

Bit 12 **MT**: Month tens in BCD format

Bits 11:8 **MU[3:0]**: Month units in BCD format

Bits 7:6 Reserved, must be kept at reset value.

Bits 5:4 **DT[1:0]**: Date tens in BCD format

Bits 3:0 **DU[3:0]**: Date units in BCD format

27.6.3 RTC control register (RTC_CR)

Address offset: 0x08

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COE	OSEL[1:0]		POL	COSEL	BKP	SUB1H	ADD1H
								rw	rw	rw	rw	rw	rw	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSIE	WUTIE	ALRBIE	ALRAIE	TSE	WUTE	ALRBE	ALRAE	Res.	FMT	BYPSHAD	REFCKON	TSEDGE	WUCKSEL[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **COE**: Calibration output enable

This bit enables the RTC_CALIB output

0: Calibration output disabled

1: Calibration output enabled

Bits 22:21 **OSEL[1:0]**: Output selection

These bits are used to select the flag to be routed to RTC_ALARM output

00: Output disabled

01: Alarm A output enabled

10: Alarm B output enabled

11: Wake-up output enabled

Bit 20 **POL**: Output polarity

This bit is used to configure the polarity of RTC_ALARM output

0: The pin is high when ALRAF/ALRBF/WUTF is asserted (depending on OSEL[1:0])

1: The pin is low when ALRAF/ALRBF/WUTF is asserted (depending on OSEL[1:0]).

Bit 19 **COSEL**: Calibration output selection

When COE=1, this bit selects which signal is output on RTC_CALIB.

0: Calibration output is 512 Hz (with default prescaler setting)

1: Calibration output is 1 Hz (with default prescaler setting)

These frequencies are valid for RTCCLK at 32.768 kHz and prescalers at their default values (PREDIV_A=127 and PREDIV_S=255). Refer to [Section 27.3.15: Calibration clock output](#)

Bit 18 **BKP**: Backup

This bit can be written by the user to memorize whether the daylight saving time change has been performed or not.

Bit 17 **SUB1H**: Subtract 1 hour (winter time change)

When this bit is set, 1 hour is subtracted to the calendar time if the current hour is not 0. This bit is always read as 0.

Setting this bit has no effect when current hour is 0.

0: No effect

1: Subtracts 1 hour to the current time. This can be used for winter time change outside initialization mode.

- Bit 16 **ADD1H**: Add 1 hour (summer time change)
 When this bit is set, 1 hour is added to the calendar time. This bit is always read as 0.
 0: No effect
 1: Adds 1 hour to the current time. This can be used for summer time change outside initialization mode.
- Bit 15 **TSIE**: Time-stamp interrupt enable
 0: Time-stamp Interrupt disable
 1: Time-stamp Interrupt enable
- Bit 14 **WUTIE**: Wake-up timer interrupt enable
 0: Wake-up timer interrupt disabled
 1: Wake-up timer interrupt enabled
- Bit 13 **ALRBIE**: Alarm B interrupt enable
 0: Alarm B Interrupt disable
 1: Alarm B Interrupt enable
- Bit 12 **ALRAIE**: Alarm A interrupt enable
 0: Alarm A interrupt disabled
 1: Alarm A interrupt enabled
- Bit 11 **TSE**: timestamp enable
 0: timestamp disable
 1: timestamp enable
- Bit 10 **WUTE**: Wake-up timer enable
 0: Wake-up timer disabled
 1: Wake-up timer enabled
Note: When the wake-up timer is disabled, wait for WUTWF=1 before enabling it again.
- Bit 9 **ALRBE**: Alarm B enable
 0: Alarm B disabled
 1: Alarm B enabled
- Bit 8 **ALRAE**: Alarm A enable
 0: Alarm A disabled
 1: Alarm A enabled
- Bit 7 Reserved, must be kept at reset value.
- Bit 6 **FMT**: Hour format
 0: 24 hour/day format
 1: AM/PM hour format
- Bit 5 **BYPHAD**: Bypass the shadow registers
 0: Calendar values (when reading from RTC_SSR, RTC_TR, and RTC_DR) are taken from the shadow registers, which are updated once every two RTCCLK cycles.
 1: Calendar values (when reading from RTC_SSR, RTC_TR, and RTC_DR) are taken directly from the calendar counters.
Note: If the frequency of the APB1 clock is less than seven times the frequency of RTCCLK, BYPSHAD must be set to '1'.

Bit 4 **REFCKON**: RTC_REFIN reference clock detection enable (50 or 60 Hz)

0: RTC_REFIN detection disabled

1: RTC_REFIN detection enabled

Note: PREDIV_S must be 0x00FF.

Bit 3 **TSEDGE**: Time-stamp event active edge

0: RTC_TS input rising edge generates a time-stamp event

1: RTC_TS input falling edge generates a time-stamp event

TSE must be reset when TSEDGE is changed to avoid unwanted TSF setting.

Bits 2:0 **WUCKSEL[2:0]**: Wake-up clock selection

000: RTC/16 clock is selected

001: RTC/8 clock is selected

010: RTC/4 clock is selected

011: RTC/2 clock is selected

10x: ck_spre (usually 1 Hz) clock is selected

11x: ck_spre (usually 1 Hz) clock is selected and 2^{16} is added to the WUT counter value
(see note below)

Note: Bits 7, 6 and 4 of this register can be written in initialization mode only (RTC_ISR/INITF = 1).

WUT = Wake-up unit counter value. WUT = (0x0000 to 0xFFFF) + 0x10000 added when WUCKSEL[2:1 = 11].

Bits 2 to 0 of this register can be written only when RTC_CR WUTE bit = 0 and RTC_ISR WUTWF bit = 1.

It is recommended not to change the hour during the calendar hour increment as it could mask the incrementation of the calendar hour.

ADD1H and SUB1H changes are effective in the next second.

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Caution: TSE must be reset when TSEDGE is changed to avoid spuriously setting of TSF.

27.6.4 RTC initialization and status register (RTC_ISR)

Address offset: 0x0C

Backup domain reset value: 0x0000 0007

System reset: not affected except INIT, INITF, and RSF bits which are cleared to '0'

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RECALPF
															r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAMP3F	TAMP2F	TAMP1F	TSOVF	TSF	WUTF	ALRBF	ALRAF	INIT	INITF	RSF	INITS	SHPF	WUTWF	ALRB WF	ALRAWF
rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rw	r	rc_w0	r	r	r	r	r

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **RECALPF**: Recalibration pending Flag

The RECALPF status flag is automatically set to '1' when software writes to the RTC_CALR register, indicating that the RTC_CALR register is blocked. When the new calibration settings are taken into account, this bit returns to '0'. Refer to [Re-calibration on-the-fly](#).

Bit 15 **TAMP3F**: RTC_TAMP3 detection flag

This flag is set by hardware when a tamper detection event is detected on the RTC_TAMP3 input.

It is cleared by software writing 0

Bit 14 **TAMP2F**: RTC_TAMP2 detection flag

This flag is set by hardware when a tamper detection event is detected on the RTC_TAMP2 input.

It is cleared by software writing 0

Bit 13 **TAMP1F**: RTC_TAMP1 detection flag

This flag is set by hardware when a tamper detection event is detected on the RTC_TAMP1 input.

It is cleared by software writing 0

Bit 12 **TSOVF**: Time-stamp overflow flag

This flag is set by hardware when a time-stamp event occurs while TSF is already set.

This flag is cleared by software by writing 0. It is recommended to check and then clear TSOVF only after clearing the TSF bit. Otherwise, an overflow might not be noticed if a time-stamp event occurs immediately before the TSF bit is cleared.

Bit 11 **TSF**: Time-stamp flag

This flag is set by hardware when a time-stamp event occurs.

This flag is cleared by software by writing 0.

Bit 10 **WUTF**: Wake-up timer flag

This flag is set by hardware when the wake-up auto-reload counter reaches 0.

This flag is cleared by software by writing 0.

This flag must be cleared by software at least 1.5 RTCCLK periods before WUTF is set to 1 again.

- Bit 9 **ALRBF**: Alarm B flag
This flag is set by hardware when the time/date registers (RTC_TR and RTC_DR) match the Alarm B register (RTC_ALRMBR).
This flag is cleared by software by writing 0.
- Bit 8 **ALRAF**: Alarm A flag
This flag is set by hardware when the time/date registers (RTC_TR and RTC_DR) match the Alarm A register (RTC_ALRMAR).
This flag is cleared by software by writing 0.
- Bit 7 **INIT**: Initialization mode
0: Free running mode
1: Initialization mode used to program time and date register (RTC_TR and RTC_DR), and prescaler register (RTC_PRER). Counters are stopped and start counting from the new value when INIT is reset.
Note: This bit is write protected. The write access procedure is described in [RTC register write protection on page 793](#).
- Bit 6 **INITF**: Initialization flag
When this bit is set to 1, the RTC is in initialization state, and the time, date and prescaler registers can be updated.
0: Calendar registers update is not allowed
1: Calendar registers update is allowed
- Bit 5 **RSF**: Registers synchronization flag
This bit is set by hardware each time the calendar registers are copied into the shadow registers (RTC_SSR, RTC_TR and RTC_DR). This bit is cleared by hardware in initialization mode, while a shift operation is pending (SHPF=1), or when in bypass shadow register mode (BYPHAD=1). This bit can also be cleared by software.
It is cleared either by software or by hardware in initialization mode.
0: Calendar shadow registers not yet synchronized
1: Calendar shadow registers synchronized
Note: This bit is write protected. The write access procedure is described in [RTC register write protection on page 793](#).
- Bit 4 **INITS**: Initialization status flag
This bit is set by hardware when the calendar year field is different from 0 (Backup domain reset state).
0: Calendar has not been initialized
1: Calendar has been initialized
- Bit 3 **SHPF**: Shift operation pending
0: No shift operation is pending
1: A shift operation is pending
This flag is set by hardware as soon as a shift operation is initiated by a write to the RTC_SHIFTR register. It is cleared by hardware when the corresponding shift operation has been executed. Writing to the SHPF bit has no effect.

Bit 2 WUTWF: Wake-up timer write flag

This bit is set by hardware up to 2 RTCCLK cycles after the WUTE bit has been set to 0 in RTC_CR, and is cleared up to 2 RTCCLK cycles after the WUTE bit has been set to 1. The wake-up timer values can be changed when WUTE bit is cleared and WUTWF is set.

0: Wake-up timer configuration update not allowed

1: Wake-up timer configuration update allowed

Bit 1 ALRBWF: Alarm B write flag

This bit is set by hardware when Alarm B values can be changed, after the ALRBE bit has been set to 0 in RTC_CR.

It is cleared by hardware in initialization mode.

0: Alarm B update not allowed

1: Alarm B update allowed

Bit 0 ALRAWF: Alarm A write flag

This bit is set by hardware when Alarm A values can be changed, after the ALRAE bit has been set to 0 in RTC_CR.

It is cleared by hardware in initialization mode.

0: Alarm A update not allowed

1: Alarm A update allowed

Note: *The bits ALRAF, ALRBF, WUTF and TSF are cleared 2 APB clock cycles after programming them to 0.*

27.6.5 RTC prescaler register (RTC_PRER)

This register must be written in initialization mode only. The initialization must be performed in two separate write accesses. Refer to [Calendar initialization and configuration on page 793](#).

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x10

Backup domain reset value: 0x007F 00FF

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREDIV_A[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	PREDIV_S[14:0]														
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **PREDIV_A[6:0]**: Asynchronous prescaler factor

This is the asynchronous division factor:

$ck_apre\ frequency = RTCCLK\ frequency / (PREDIV_A + 1)$

Bit 15 Reserved, must be kept at reset value.

Bits 14:0 **PREDIV_S[14:0]**: Synchronous prescaler factor

This is the synchronous division factor:

$ck_spre\ frequency = ck_apre\ frequency / (PREDIV_S + 1)$

27.6.6 RTC wake-up timer register (RTC_WUTR)

This register can be written only when WUTWF is set to 1 in RTC_ISR.

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x14

Backup domain reset value: 0x0000 FFFF

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WUT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **WUT[15:0]**: Wake-up auto-reload value bits

When the wake-up timer is enabled (WUTE set to 1), the WUTF flag is set every (WUT[15:0] + 1) ck_wut cycles. The ck_wut period is selected through WUCKSEL[2:0] bits of the RTC_CR register

When WUCKSEL[2] = 1, the wake-up timer becomes 17-bits and WUCKSEL[1] effectively becomes WUT[16] the most-significant bit to be reloaded into the timer.

The first assertion of WUTF occurs (WUT+1) ck_wut cycles after WUTE is set. Setting WUT[15:0] to 0x0000 with WUCKSEL[2:0] = 011 (RTCCLK/2) is forbidden.

27.6.7 RTC alarm A register (RTC_ALRMAR)

This register can be written only when ALRAWF is set to 1 in RTC_ISR, or in initialization mode.

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x1C

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MSK4	WDSEL	DT[1:0]		DU[3:0]				MSK3	PM	HT[1:0]		HU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MSK2	MNT[2:0]			MNU[3:0]				MSK1	ST[2:0]			SU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **MSK4**: Alarm A date mask

- 0: Alarm A set if the date/day match
- 1: Date/day don't care in Alarm A comparison

Bit 30 **WDSEL**: Week day selection

- 0: DU[3:0] represents the date units
- 1: DU[3:0] represents the week day. DT[1:0] is don't care.

Bits 29:28 **DT[1:0]**: Date tens in BCD format.

Bits 27:24 **DU[3:0]**: Date units or day in BCD format.

Bit 23 **MSK3**: Alarm A hours mask

- 0: Alarm A set if the hours match
- 1: Hours don't care in Alarm A comparison

Bit 22 **PM**: AM/PM notation

- 0: AM or 24-hour format
- 1: PM

Bits 21:20 **HT[1:0]**: Hour tens in BCD format.

Bits 19:16 **HU[3:0]**: Hour units in BCD format.

Bit 15 **MSK2**: Alarm A minutes mask

- 0: Alarm A set if the minutes match
- 1: Minutes don't care in Alarm A comparison

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format.

Bits 11:8 **MNU[3:0]**: Minute units in BCD format.

Bit 7 **MSK1**: Alarm A seconds mask

- 0: Alarm A set if the seconds match
- 1: Seconds don't care in Alarm A comparison

Bits 6:4 **ST[2:0]**: Second tens in BCD format.

Bits 3:0 **SU[3:0]**: Second units in BCD format.

27.6.8 RTC alarm B register (RTC_ALRMBR)

This register can be written only when ALRBWF is set to 1 in RTC_ISR, or in initialization mode.

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x20

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MSK4	WDSEL	DT[1:0]		DU[3:0]				MSK3	PM	HT[1:0]		HU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MSK2	MNT[2:0]			MNU[3:0]				MSK1	ST[2:0]			SU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **MSK4**: Alarm B date mask

0: Alarm B set if the date and day match

1: Date and day don't care in Alarm B comparison

Bit 30 **WDSEL**: Week day selection

0: DU[3:0] represents the date units

1: DU[3:0] represents the week day. DT[1:0] is don't care.

Bits 29:28 **DT[1:0]**: Date tens in BCD format

Bits 27:24 **DU[3:0]**: Date units or day in BCD format

Bit 23 **MSK3**: Alarm B hours mask

0: Alarm B set if the hours match

1: Hours don't care in Alarm B comparison

Bit 22 **PM**: AM/PM notation

0: AM or 24-hour format

1: PM

Bits 21:20 **HT[1:0]**: Hour tens in BCD format

Bits 19:16 **HU[3:0]**: Hour units in BCD format

Bit 15 **MSK2**: Alarm B minutes mask

0: Alarm B set if the minutes match

1: Minutes don't care in Alarm B comparison

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format

Bits 11:8 **MNU[3:0]**: Minute units in BCD format

Bit 7 **MSK1**: Alarm B seconds mask

0: Alarm B set if the seconds match

1: Seconds don't care in Alarm B comparison

Bits 6:4 **ST[2:0]**: Second tens in BCD format

Bits 3:0 **SU[3:0]**: Second units in BCD format

27.6.9 RTC write protection register (RTC_WPR)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	KEY[7:0]							
								w	w	w	w	w	w	w	w

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **KEY[7:0]**: Write protection key

This byte is written by software.

Reading this byte always returns 0x00.

Refer to [RTC register write protection](#) for a description of how to unlock RTC register write protection.

27.6.10 RTC sub second register (RTC_SSR)

Address offset: 0x28

Backup domain reset value: 0x0000 0000

System reset: 0x0000 0000 when BYPSHAD = 0. Not affected when BYPSHAD = 1.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SS[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **SS[15:0]**: Sub second value

SS[15:0] is the value in the synchronous prescaler counter. The fraction of a second is given by the formula below:

Second fraction = (PREDIV_S - SS) / (PREDIV_S + 1)

Note: SS can be larger than PREDIV_S only after a shift operation. In that case, the correct time/date is one second less than as indicated by RTC_TR/RTC_DR.

27.6.11 RTC shift control register (RTC_SHIFTR)

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x2C

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADD1S	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
w															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SUBFS[14:0]														
	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bit 31 **ADD1S**: Add one second

0: No effect

1: Add one second to the clock/calendar

This bit is write only and is always read as zero. Writing to this bit has no effect when a shift operation is pending (when SHPF=1, in RTC_ISR).

This function is intended to be used with SUBFS (see description below) in order to effectively add a fraction of a second to the clock in an atomic operation.

Bits 30:15 Reserved, must be kept at reset value.

Bits 14:0 **SUBFS[14:0]**: Subtract a fraction of a second

These bits are write only and is always read as zero. Writing to this bit has no effect when a shift operation is pending (when SHPF=1, in RTC_ISR).

The value which is written to SUBFS is added to the synchronous prescaler counter. Since this counter counts down, this operation effectively subtracts from (delays) the clock by:

Delay (seconds) = SUBFS / (PREDIV_S + 1)

A fraction of a second can effectively be added to the clock (advancing the clock) when the ADD1S function is used in conjunction with SUBFS, effectively advancing the clock by:

Advance (seconds) = (1 - (SUBFS / (PREDIV_S + 1))).

Note: Writing to SUBFS causes RSF to be cleared. Software can then wait until RSF=1 to be sure that the shadow registers have been updated with the shifted time.

27.6.12 RTC timestamp time register (RTC_TSTR)

The content of this register is valid only when TSF is set to 1 in RTC_ISR. It is cleared when TSF bit is reset.

Address offset: 0x30

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]			
									r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]			
	r	r	r	r	r	r	r		r	r	r	r	r	r	r

Bits 31:23 Reserved, must be kept at reset value.

Bit 22 **PM**: AM/PM notation

0: AM or 24-hour format

1: PM

Bits 21:20 **HT[1:0]**: Hour tens in BCD format.

Bits 19:16 **HU[3:0]**: Hour units in BCD format.

Bit 15 Reserved, must be kept at reset value.

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format.

Bits 11:8 **MNU[3:0]**: Minute units in BCD format.

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **ST[2:0]**: Second tens in BCD format.

Bits 3:0 **SU[3:0]**: Second units in BCD format.

27.6.13 RTC timestamp date register (RTC_TSDR)

The content of this register is valid only when TSF is set to 1 in RTC_ISR. It is cleared when TSF bit is reset.

Address offset: 0x34

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDU[2:0]			MT	MU[3:0]				Res.	Res.	DT[1:0]		DU[3:0]			
r	r	r	r	r	r	r	r			r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:13 **WDU[2:0]**: Week day units

Bit 12 **MT**: Month tens in BCD format

Bits 11:8 **MU[3:0]**: Month units in BCD format

Bits 7:6 Reserved, must be kept at reset value.

Bits 5:4 **DT[1:0]**: Date tens in BCD format

Bits 3:0 **DU[3:0]**: Date units in BCD format

27.6.14 RTC time-stamp sub second register (RTC_TSSSR)

The content of this register is valid only when RTC_ISR/TSF is set. It is cleared when the RTC_ISR/TSF bit is reset.

Address offset: 0x38

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SS[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **SS[15:0]**: Sub second value

SS[15:0] is the value of the synchronous prescaler counter when the timestamp event occurred.

27.6.15 RTC calibration register (RTC_CALR)

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#).

Address offset: 0x3C

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CALP	CALW8	CALW16	Res.	Res.	Res.	Res.	CALM[8:0]								
rw	rw	rw					rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **CALP**: Increase frequency of RTC by 488.5 ppm

0: No RTCCLK pulses are added.

1: One RTCCLK pulse is effectively inserted every 2^{11} pulses (frequency increased by 488.5 ppm).

This feature is intended to be used in conjunction with CALM, which lowers the frequency of the calendar with a fine resolution. If the input frequency is 32768 Hz, the number of RTCCLK pulses added during a 32-second window is calculated as follows: $(512 * CALP) - CALM$.

Refer to [Section 27.3.12: RTC smooth digital calibration](#).

Bit 14 **CALW8**: Use an 8-second calibration cycle period

When CALW8 is set to '1', the 8-second calibration cycle period is selected.

Note: CALM[1:0] are stuck at "00" when CALW8='1'. Refer to [Section 27.3.12: RTC smooth digital calibration](#).

Bit 13 **CALW16**: Use a 16-second calibration cycle period

When CALW16 is set to '1', the 16-second calibration cycle period is selected. This bit must not be set to '1' if CALW8=1.

Note: CALM[0] is stuck at '0' when CALW16='1'. Refer to [Section 27.3.12: RTC smooth digital calibration](#).

Bits 12:9 Reserved, must be kept at reset value.

Bits 8:0 **CALM[8:0]**: Calibration minus

The frequency of the calendar is reduced by masking CALM out of 2^{20} RTCCLK pulses (32 seconds if the input frequency is 32768 Hz). This decreases the frequency of the calendar with a resolution of 0.9537 ppm.

To increase the frequency of the calendar, this feature should be used in conjunction with CALP. See [Section 27.3.12: RTC smooth digital calibration on page 797](#).

27.6.16 RTC tamper and alternate function configuration register (RTC_TAFCR)

Address offset: 0x40

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PC15 MODE	PC15 VALUE	PC14 MODE	PC14 VALUE	PC13 MODE	PC13 VALUE	Res.	Res.
								r/w	r/w	r/w	r/w	r/w	r/w		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAMPP UDIS	TAMPPRCH [1:0]		TAMPFLT[1:0]		TAMPFREQ[2:0]			TAMPT S	TAMP3 TRG ⁽¹⁾	TAMP3 E ⁽¹⁾	TAMP2 TRG	TAMP2 E	TAMPIE	TAMP1 TRG	TAMP1 E
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

1. RTC_TAMP3 is not available on the STM32F303x6/8 and STM32F328 devices.

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **PC15MODE**: PC15 mode

0: PC15 is controlled by the GPIO configuration registers. Consequently PC15 is floating in Standby mode.

1: PC15 is forced to push-pull output if LSE is disabled.

Bit 22 **PC15VALUE**: PC15 value

If the LSE is disabled and PC15MODE = 1, PC15VALUE configures the PC15 output data.

Bit 21 **PC14MODE**: PC14 mode

0: PC14 is controlled by the GPIO configuration registers. Consequently PC14 is floating in Standby mode.

1: PC14 is forced to push-pull output if LSE is disabled.

Bit 20 **PC14VALUE**: PC14 value

If the LSE is disabled and PC14MODE = 1, PC14VALUE configures the PC14 output data.

Bit 19 **PC13MODE**: PC13 mode

0: PC13 is controlled by the GPIO configuration registers. Consequently PC13 is floating in Standby mode.

1: PC13 is forced to push-pull output if all RTC functions are disabled.

Bit 18 **PC13VALUE**: RTC_ALARM output type/PC13 value

If PC13 is used to output RTC_ALARM, PC13VALUE configures the output configuration:

0: RTC_ALARM is an open-drain output

1: RTC_ALARM is a push-pull output

If all RTC functions are disabled and PC13MODE = 1, PC13VALUE configures the PC13 output data.

Bits 17:16 Reserved, must be kept at reset value.

Bit 15 **TAMPPUDIS**: RTC_TAMPx pull-up disable

This bit determines if each of the RTC_TAMPx pins are pre-charged before each sample.

0: Precharge RTC_TAMPx pins before sampling (enable internal pull-up)

1: Disable precharge of RTC_TAMPx pins.

Bits 14:13 **TAMPPRCH[1:0]**: RTC_TAMPx precharge duration

These bits determine the duration of time during which the pull-up is activated before each sample. TAMPPRCH is valid for each of the RTC_TAMPx inputs.

- 0x0: 1 RTCCLK cycle
- 0x1: 2 RTCCLK cycles
- 0x2: 4 RTCCLK cycles
- 0x3: 8 RTCCLK cycles

Bits 12:11 **TAMPFLT[1:0]**: RTC_TAMPx filter count

These bits determine the number of consecutive samples at the specified level (TAMP*TRG) needed to activate a Tamper event. TAMPFLT is valid for each of the RTC_TAMPx inputs.

- 0x0: Tamper event is activated on edge of RTC_TAMPx input transitions to the active level (no internal pull-up on RTC_TAMPx input).
- 0x1: Tamper event is activated after 2 consecutive samples at the active level.
- 0x2: Tamper event is activated after 4 consecutive samples at the active level.
- 0x3: Tamper event is activated after 8 consecutive samples at the active level.

Bits 10:8 **TAMPFREQ[2:0]**: Tamper sampling frequency

Determines the frequency at which each of the RTC_TAMPx inputs are sampled.

- 0x0: RTCCLK / 32768 (1 Hz when RTCCLK = 32768 Hz)
- 0x1: RTCCLK / 16384 (2 Hz when RTCCLK = 32768 Hz)
- 0x2: RTCCLK / 8192 (4 Hz when RTCCLK = 32768 Hz)
- 0x3: RTCCLK / 4096 (8 Hz when RTCCLK = 32768 Hz)
- 0x4: RTCCLK / 2048 (16 Hz when RTCCLK = 32768 Hz)
- 0x5: RTCCLK / 1024 (32 Hz when RTCCLK = 32768 Hz)
- 0x6: RTCCLK / 512 (64 Hz when RTCCLK = 32768 Hz)
- 0x7: RTCCLK / 256 (128 Hz when RTCCLK = 32768 Hz)

Bit 7 **TAMPTS**: Activate timestamp on tamper detection event

- 0: Tamper detection event does not cause a timestamp to be saved
- 1: Save timestamp on tamper detection event

TAMPTS is valid even if TSE=0 in the RTC_CR register.

Bit 6 **TAMP3TRG**: Active level for RTC_TAMP3 input

- if TAMPFLT != 00:
 - 0: RTC_TAMP3 input staying low triggers a tamper detection event.
 - 1: RTC_TAMP3 input staying high triggers a tamper detection event.
- if TAMPFLT = 00:
 - 0: RTC_TAMP3 input rising edge triggers a tamper detection event.
 - 1: RTC_TAMP3 input falling edge triggers a tamper detection event.

Bit 5 **TAMP3E**: RTC_TAMP3 detection enable

- 0: RTC_TAMP3 input detection disabled
- 1: RTC_TAMP3 input detection enabled

Bit 4 **TAMP2TRG**: Active level for RTC_TAMP2 input

- if TAMPFLT != 00:
 - 0: RTC_TAMP2 input staying low triggers a tamper detection event.
 - 1: RTC_TAMP2 input staying high triggers a tamper detection event.
- if TAMPFLT = 00:
 - 0: RTC_TAMP2 input rising edge triggers a tamper detection event.
 - 1: RTC_TAMP2 input falling edge triggers a tamper detection event.

- Bit 3 **TAMP2E**: RTC_TAMP2 input detection enable
0: RTC_TAMP2 detection disabled
1: RTC_TAMP2 detection enabled
- Bit 2 **TAMPIE**: Tamper interrupt enable
0: Tamper interrupt disabled
1: Tamper interrupt enabled.
- Bit 1 **TAMP1TRG**: Active level for RTC_TAMP1 input
If TAMPFLT != 00
0: RTC_TAMP1 input staying low triggers a tamper detection event.
1: RTC_TAMP1 input staying high triggers a tamper detection event.
if TAMPFLT = 00:
0: RTC_TAMP1 input rising edge triggers a tamper detection event.
1: RTC_TAMP1 input falling edge triggers a tamper detection event.
- Bit 0 **TAMP1E**: RTC_TAMP1 input detection enable
0: RTC_TAMP1 detection disabled
1: RTC_TAMP1 detection enabled

Caution: When TAMPFLT = 0, TAMPxE must be reset when TAMPxTRG is changed to avoid spuriously setting TAMPxF.

27.6.17 RTC alarm A sub second register (RTC_ALRMASR)

This register can be written only when ALRAE is reset in RTC_CR register, or in initialization mode.

This register is write protected. The write access procedure is described in [RTC register write protection on page 793](#)

Address offset: 0x44

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	MASKSS[3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
				rw	rw	rw	rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SS[14:0]														
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	w	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 MASKSS[3:0]: Mask the most-significant bits starting at this bit

0: No comparison on sub seconds for Alarm A. The alarm is set when the seconds unit is incremented (assuming that the rest of the fields match).

1: SS[14:1] are don't care in Alarm A comparison. Only SS[0] is compared.

2: SS[14:2] are don't care in Alarm A comparison. Only SS[1:0] are compared.

3: SS[14:3] are don't care in Alarm A comparison. Only SS[2:0] are compared.

...

12: SS[14:12] are don't care in Alarm A comparison. SS[11:0] are compared.

13: SS[14:13] are don't care in Alarm A comparison. SS[12:0] are compared.

14: SS[14] is don't care in Alarm A comparison. SS[13:0] are compared.

15: All 15 SS bits are compared and must match to activate alarm.

The overflow bits of the synchronous counter (bits 15) is never compared. This bit can be different from 0 only after a shift operation.

Bits 23:15 Reserved, must be kept at reset value.

Bits 14:0 SS[14:0]: Sub seconds value

This value is compared with the contents of the synchronous prescaler counter to determine if Alarm A is to be activated. Only bits 0 up MASKSS-1 are compared.

27.6.18 RTC alarm B sub second register (RTC_ALRMBSSR)

This register can be written only when ALRBE is reset in RTC_CR register, or in initialization mode.

This register is write protected. The write access procedure is described in [Section : RTC register write protection](#).

Address offset: 0x48

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	MASKSS[3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
				r/w	r/w	r/w	r/w								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SS[14:0]														
	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	w	r/w	r/w

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **MASKSS[3:0]**: Mask the most-significant bits starting at this bit

0x0: No comparison on sub seconds for Alarm B. The alarm is set when the seconds unit is incremented (assuming that the rest of the fields match).

0x1: SS[14:1] are don't care in Alarm B comparison. Only SS[0] is compared.

0x2: SS[14:2] are don't care in Alarm B comparison. Only SS[1:0] are compared.

0x3: SS[14:3] are don't care in Alarm B comparison. Only SS[2:0] are compared.

...

0xC: SS[14:12] are don't care in Alarm B comparison. SS[11:0] are compared.

0xD: SS[14:13] are don't care in Alarm B comparison. SS[12:0] are compared.

0xE: SS[14] is don't care in Alarm B comparison. SS[13:0] are compared.

0xF: All 15 SS bits are compared and must match to activate alarm.

The overflow bits of the synchronous counter (bits 15) is never compared. This bit can be different from 0 only after a shift operation.

Bits 23:15 Reserved, must be kept at reset value.

Bits 14:0 **SS[14:0]**: Sub seconds value

This value is compared with the contents of the synchronous prescaler counter to determine if Alarm B is to be activated. Only bits 0 up to MASKSS-1 are compared.

27.6.19 RTC backup registers (RTC_BKPxR)

Address offset: 0x50 to 0x8C

Backup domain reset value: 0x0000 0000

System reset: not affected

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BKP[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BKP[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	W	rW	rW

Bits 31:0 BKP[31:0]

The application can write or read data to and from these registers.

They are powered-on by V_{BAT} when V_{DD} is switched off, so that they are not reset by

System reset, and their contents remain valid when the device operates in low-power mode.

This register is reset on a tamper detection event, as long as TAMPxF=1.

27.6.20 RTC register map

Table 141. RTC register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	RTC_TR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]				Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]			
	Reset value										0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x04	RTC_DR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	YT[3:0]				YU[3:0]				WDU[2:0]			MT	MU[3:0]				Res.	Res.	DT[1:0]		DU[3:0]			
	Reset value									0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1		0	0	0	0	0	1
0x08	RTC_CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COE	OSEL[1:0]		POL	COSEL	BKP	SUB1H	ADD1H	TSIE	WUTIE	ALRBIE	ALRAIE	TSE	WUTE	ALRBE	ALRAE	Res.	FMT	BYPHAD	REFCKON	TSEDGE	WUCKSEL[2:0]		
	Reset value									0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	RTC_ISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RECALPF	TAMP3F	TAMP2F	TAMP1F	TSOVF	TSF	WUTF	ALRBF	ALRAF	INIT	INITF	RSF	INTS	SHPF	WUTF	ALRWF	ALRAWF
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
0x10	RTC_PRER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREDIV_A[6:0]							PREDIV_S[14:0]																
	Reset value									1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
0x14	RTC_WUTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WUT[15:0]															
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 141. RTC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x1C	RTC_ALRMAR	MSK4	WDSEL	DT[1:0]		DU[3:0]				MSK3	PM	HT[1:0]		HU[3:0]				MSK2	MNT[2:0]				MNU[3:0]				MSK1	ST[2:0]			SU[3:0]								
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x20	RTC_ALRMBR	MSK4	WDSEL	DT[1:0]		DU[3:0]				MSK3	PM	HT[1:0]		HU[3:0]				MSK2	MNT[2:0]				MNU[3:0]				MSK2	ST[2:0]			SU[3:0]								
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x24	RTC_WPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	KEY													
	Reset value																									0	0	0	0	0	0	0	0	0					
0x28	RTC_SSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SS[15:0]																					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x2C	RTC_SHIFTR	ADD1S	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUBFS[14:0]																				
	Reset value	0																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x30	RTC_TSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]				Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]									
	Reset value										0	0	0	0	0	0	0		0	0	0	0	0	0	0		0	0	0	0	0	0	0	0					
0x34	RTC_TSDR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WDU[1:0]			MT	MU[3:0]				Res.	Res.	DT[1:0]			DU[3:0]								
	Reset value																	0	0	0	0	0	0	0	0			0	0	0	0	0	0	0					
0x38	RTC_TSSSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SS[15:0]																					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x3C	RTC_CALR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CALP	CALW8	CALW16	Res.	Res.	Res.	Res.	CALM[8:0]														
	Reset value																	0	0	0					0	0	0	0	0	0	0	0	0	0					
0x40	RTC_TAFCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PC15MODE	PC15VALUE	PC14MODE	PC14VALUE	PC13MODE	PC13VALUE	Res.	Res.	TAMPPUDIS	TAMPPRCH[1:0]			TAMPFLT[1:0]			TAMPFREQ[2:0]			TAMPTS	TAMP3TRG	TAMP3E	TAMP2TRG	TAMP2E	TAMPIE	TAMP1TRG	TAMP1E				
	Reset value									0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x44	RTC_ALRMASSR	Res.	Res.	Res.	Res.	MASKSS [3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SS[14:0]																				
	Reset value					0	0	0	0										0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x48	RTC_ALRMBSSR	Res.	Res.	Res.	Res.	MASKSS [3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SS[14:0]																				
	Reset value					0	0	0	0										0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

Table 141. RTC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x4C	RTC_OR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RTC_ALARM_TYPE	TSINSEL[1:0]		0
	Reset value																													0	0	0	
0x50 to 0x8C	RTC_BKP0R	BKP[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	to RTC_BKP15R	BKP[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.